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Enhancing Transparency in Student Performance Prediction Using Explainable Artificial Intelligence (XAI)

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Abstract

Determining in advance student performance is necessary for providing early intervention and enhancing academic results. To gain interpretable insights in machine learning models used to forecast student performance, this research uses SHAP and LIME based Explainable Artificial Intelligence (XAI) techniques. The study develops Machine Learning models based on Random Forest, Support Vector Machines (SVM) and Multi-Layer Perceptron (MLP) using the Student Performance dataset. The results from the Random Forest model had the highest accuracy of 83.07% and the highest F1 score of 0.905. By providing global and local explanatory power of predictions, stakeholder can understand why a given student can do well and why a failure can occur using SHAP and LIME explanations. The overall goal is to provide evidence that XAI in these systems can improve transparency, trust, and ethical decision making in the student support and policy development processes.

1 Introduction

Technology is changing the face of the changing education of today. Nowadays modern tools and software are being used by schools and colleges so that students can do better. Predicting how the students will perform academically is one powerful way helping the students. If teachers can aid students before it is too late, by predicting early enough, which students are likely to struggle, we can save them from dropping out of the program altogether (Romero & Ventura, 2020). There are many ways a student cannot perform in school. Family background, health, motivation, study habits and even how often they attend classes are all included in these. All students have their own strengths and weaknesses (Wang et al., 2022). Nevertheless, traditional schools as a rule treat students similarly. For teaching and supporting everyone, they utilize the same methods. This “one size fits all” doesn’t work for a lot of students. Data is being used in schools to solve this problem. In addition, these methods utilize data about the students for creating a better understanding of their needs. Another example would be, by looking at the student's past grades, attendance records, and family situations, teachers can determine who requires an extra boost. One technology that helps schools to do this very effectively is machine learning (ML). Prediction with ML is based on trying to use past data to predict future outcomes. Many ML models are, however, complicated. Finally, teachers and students do not understand how these models make those predictions (Ribeiro et al., 2016). If teachers do not understand why a prediction was made, they may not trust it. As such, they are difficult to use for making predictions in real life. That is why it is necessary to make prediction easy to understand.

1.2 Problem Statement

There are many machine learning models that predict the student performance accurately. However, many of these models are 'black boxes. Results are produced in a black box without showing how those results were decided. Problems result, however, because teachers must know the reasons some students are at risk. With that, teachers may doubt the results or refuse to use them, without having a clue to the reasons. It only makes the question of fairness regarding what actually happens in the air more important. For instance, if the predictions are ambiguous, it may potentially result in unfairness of some students to some groups (Hooshyar and Yang, 2024).

To make better decision, models explaining schools’ predictions need to be constructed. Teachers are provided with transparent predictions to help them to trust and to use the predictions more effectively.

1.3 Aim of the Research

The main aim of this project is to augment how well machine learning models describe their predictions of student performance. More specifically, these methods from Explainable Artificial Intelligence (XAI) are used in this project; they are SHAP and LIME. They make it clear how and why the models get to the prediction they do.

1.4 Research Objectives

The project will achieve the main goal by completing four clear objectives:

Prediction of student academic performance is developed using machine learning models (Random Forest, Support Vector Machines, and Multi-Layer Perceptron) and implemented. LIME and SHAP are used to explain these models in a clear, easy to understand manner. You will know which factors like attendance, family support or study habits are the most important to predict the student success or failure.

Explain the roles that clear explanations can play for teachers in detecting at risk students early and intervening.

1.5 Research Question

The main question that guides this research is:

"How can Explainable Artificial Intelligence (XAI) techniques like SHAP and LIME be effectively used to make accurate predictions about student performance, while clearly explaining the socio-demographic, behavioural, and motivational reasons behind these predictions?"

1.6 Key Contributions

This research makes several important contributions to education and technology like it uses SHAP and LIME to make the predictions understandable at both group and individual student level, it generates new features from the data (i.e. study efficiency, level of commitments, available supports). It also considers three different machine learning models in its testing and contrast under the environment of balanced and imbalanced data. Finally, fairness and transparency of the prediction methods are also included in it.

1.7 Why This Research Matters

Clear understandable predictions help the teachers and schools act quickly. Support at the early stage can help the students from getting failed or out of the school. Such explainable predictions also create trust. Teachers are also more likely to use technology if they can make the reasons for each prediction clear to the class. Furthermore, clear predictions aid impeding unfairness from stemming from the model by making it easy to identify any biases in the model (Ribeiro et al., 2016).

1.8 Structure of the Report

This report includes six chapters, each of them clearly explaining a different part of this research. Chapter 2 presents the literature review, which explores the past relevant studies on student performance prediction and explainable AI and identifies the gaps that this study aims to address. Chapter 3 outlines the methodology, describing the dataset used, the steps taken to prepare the data, the creation of new features, machine learning models selected, and the explainability tools applied, SHAP and LIME. Chapter 4 focuses on implementation and results, providing clear and detailed explanations of how the models were trained, tuned, and evaluated, and presenting the outcomes of applying SHAP and LIME. Chapter 5 analysis and discussion, interpreting the results in terms of their relevance to educational settings and examining issues such as fairness, transparency, and ethical use. Finally, Chapter 6 concludes the report by summarizing all the key findings and suggesting directions for future research and development.

2 Literature Review

Research around predictive modelling in education has been a long-standing focus and it stems from the need to find tools of data driven learners at high risk for academic failure and to enhance learning outcomes. Section 2.1 reviews prior work on student performance prediction models and Section 2.2 discusses challenges regarding interpretability and transparency that has motivated the union of XAI. Section 2.3 introduces the specific XAI techniques of SHAP and LIME, and Section 2.4 demonstrates how these methodologies have been used in the education domain. In the last section of this thesis (2.5), we pinpoint the gaps in the current body of literature that this thesis hopes to cure.

Many of these studies have improved the understanding of student performance prediction, notwithstanding that most of them aim at high accuracy, to the detriment of model clarity. Also, there is a gap in application of Explainable AI tool such as SHAP and LIME in an already built structured feature added ML pipeline, such as secondary school level. Previous models that exist in the literature are built upon, with more emphasis on interpretability and fairness, still not being deeply explored.

2.1 Overview of Student Performance Prediction Models

Student Performance Prediction (SPP) refers to the task of forecasting academic outcomes—such as course grades, GPA, or dropout likelihood—using historical, behavioural, or demographic data. In the early 2000s, previous approaches for SPP were relatively simple machine learning models and small sized feature sets. For example, in Kotsiantis et al. (2004) they used instance of decision trees, neural networks, and logistic regression to predict several simple predictors related to assignment and quiz performance and even basic algorithms such as those can outperform random guessing by leveraging student's pattern. The field has ended up embracing not just traditional, but also the state of the art approaches to ML. Albreiki et al. (2021), Romero and Ventura (2020), as well as several other sources, list a long list of common algorithms: Decision trees and Naive Bayes (easy), Support Vector Machines, Ensemble methods like Random Forests and Gradient Boosted Trees, and even newer such as deep neural networks or hybrid model. A variety of input features based on aspect of a student's profile can be handled by these models. An early benchmark study based on Portuguese secondary schools' data for predicting students' final grades in Math and Portuguese courses was offered by Cortez and Silva (2008) This project also uses the Cortez and Silva dataset, not only for its historical relevance, but to extend their approach through modern feature engineering and explanation tools that were unavailable at the time. Decision tree algorithms were applied to the students' socio demographic and academic features, and this was shown to be possible to identify at risk students with reasonable accuracy. Other researchers have since searched for an ensemble approach that combines several models for better accuracy in subsequent years. One example is Albreiki et al. (2021) examines all work in the literature concerning SPP and stated that often ensemble models (e.g., Random Forests or voting classifiers) can outperform single models because they provide the capability to encompass a wider set of patterns. According to the combination of feature analysis and machine learning, Lu et al. (2024) tried to investigate factors that influence college students' performances and indicated that ensemble methods and hybrid techniques could achieve high accuracy via combining the complementary strengths of various algorithms.

2.2 Challenges in Interpretability and Transparency

Production deployment of predictive systems in educational institutions demands both predictive accuracy and interpretable capabilities. Educational professionals need complete explanations from models to both confirm the validity of outputs and support decision-making processes that lead to informed actions (Ribeiro et al., 2016). The primary challenges include: The operational black-box nature of neural networks along with ensemble models fails to provide transparency about which input features contribute to predictions according to (Hooshyar and Yang 2024). Members of stakeholder communities show reluctance to deploy or support artificial intelligence programs which they do not comprehend as they make decisions especially when they work in admission processes and graduation certification systems and disciplinary methods (Al-Alawi et al., 2023). Models receive training data containing hidden biases which cause unfavourable treatment of specific student populations. Such biases cannot be detected because interpretability is missing from these systems (Alsariera et al., 2022). Educators and administrators encounter difficulties when attempting to correctly understand explanations even when these explanations are supplied to them. The requirement exists for understandable explanations that can guide appropriate decisions.

2.3 Explainable AI Techniques (SHAP and LIME)

SHAP and LIME provide reliable explanations that do not weaken existing ML sources, therefore maintaining the mechanisms of original predictive frameworks.

The feature importance assessment tool SHAP delivers predictions through Shapley value assignments to determine what extent each attribute affects the result (Lundberg and Lee, 2017). This approach brings together consistent and location-specific accuracy when explaining predictions. A prediction of student failure allows SHAP to generate numerical values showing the percentage contribution from each factor including poor attendance and low prior grades. SHAP offers dual functionality for both point-wise explanations of specific predictions and universal explanation of feature rankings throughout the entire dataset.

LIME constructs local surrogate models that explain individual predictions through controlled changes within the data which trigger changes in output results (Ribeiro et al., 2016). The algorithm determines a plain model like linear regression that properly represents the complicated model within its proximity. The local explanation feature of this method helps reveal individual predictions such as drop-out risk for specific students.

Both methods have been effectively used in education to enhance understanding and foster trust in AI models.

2.4 Case Studies and Applications in Education

Several empirical studies illustrate the successful application of SHAP and LIME to student performance prediction, like LIME, which has also been used by Dib and Capus (2025) to predict and explain dropout risks in an intelligent tutoring system.

When the model marked a student as at risk of dropping out in their study, LIME was used to explain the model's prediction of that student. LIME's transparency enabled teachers and mentors to learn about the warning signs, or indicators (in this case, poor homework scores or low quiz participation), earlier and provide intervention to these students.

Zafari et al. (2021) and Aguirre Montaña and Cabrera Loayza (2020) demonstrated that a combination of machine learning and explanation methods can enhance early prediction of the academic outcomes and dropout by predicting early in a term how well a student is doing and explaining these predictions, so educators have a valuable look into action before it becomes too late. For example, when a model suggests that a student early on likely will fail and flags lack of forum participation as a primary influential factor, an instructor can step in and suggest to that student to be more active on the forum.

In Wang et al. (2022), interpretability was emphasised in their findings in the analysis of factors related to college student performance with the use of machine learning. With XAI tools, they were sure of the results of their analysis (such as which factors most influence GPA), in other words, transparency. This validated that the model was finding these important features (like studying or attending classes) that are meaningful and can be explained with respect to educational theory.

SHAP and LIME are used both by George et al. (2025) and Johora et al. (2025) to identify influential factors such as GPA, attendance, and midterm performance. This allowed them to determine which factors were responsible for student success or failure. This gave us an insight into the level we did not have previously, and we could develop targeted interventions like reaching out to students with low attendance before their performance degrades. SHAP and LIME work together to provide a fuller picture—the class (global explanation) and the individual student (local explanation)—that educators can work to resolve at that first level. Abinaya et al. (2023) and Elsaid Khoudier et al. (2023) also made use of SHAP and LIME for the same purpose of classifying students into different characters of academic achievement levels. It turned out that the explainability provided by such tools was very important for deriving reasonable educational policies. Or put another way, it isn't good enough to know who high or low achievers may be, knowing 'why' (for example, because of attendance, background, or study habits) enables school administrators to develop ways to make policies or develop support programs for those specific causes.

The strengths of these case studies are taken together and used to accomplish a combination of prediction and explanation. Nevertheless, this work differs from most of the other works as it combines feature engineering, interpretable models, and detailed case level explanations. This dual use of SHAP and LIME also further enriches both the patterns at the group level and student profiles at the individual level to enable technical outputs to match educational needs.

2.5 Gaps in Existing Literature

Explaining AI (XAI) has been studied in many ways in education and many of these studies were in relation with college or online learning in specific areas — especially in the West. These models and explanation tools work very different in other places or for other types of students. For example, most research about XAI in student performance prediction (SPP) has been performed in Western universities or online platforms. These methods should be tested in K-12 schools, job training programs and in schools in other countries, outside the West. It will allow us to once test whether the methods are usable for other students and cultures. To address this issue this study, makes use of data from a public secondary school then develops models for such education levels. Nevertheless, more testing in other types of schools is needed. A further problem is that there are no long-term studies. On the other hand, many researchers examine just one moment in time to see what their prediction of student

performance is like only once. Very few research follows students over the years to see the true effect of XAI tools. A good example would follow the trajectories of students who were assisted by explainable models in their first and then count whether these students get better grades by graduation. But we don't know how well the models perform over time because students change, or school systems change. Future studies should further analyse how these are performing in many school terms or school years. Research studies lack investigations into the usability perceptions that teachers and students together with school staff assign to these explanation tools. There are no verified findings about how teachers interpret SHAP plots or LIME explanations and whether those tools improve their judgment quality. These explanation tools could possibly introduce additional complications for users. The development process of tools neglects user needs and additional testing for usability remains scarce. Academic work needs to study user opinions more and find better ways to make explanations easier for people to comprehend. Explanations should include basic images along with common school vocabulary for better understanding. The majority of educational XAI investigations utilize outdated data collected from offline testing environments. Learning applications and educational software that function in actual student environments require instant prediction responses together with explanations that appear while students actively use the systems. The system needs the capability to instantly provide answers in real time when a teacher asks questions about possible alterations in student study time. This kind of interactive and fast XAI remains a new and challenging concept to create in the present day. Building user-friendly tools at high speeds presents the main technical obstacle.

Finally, the decision between using basic understandable models versus advanced accurate models requires analysis of additional interpretive tools. The applicability of post-hoc explanation tools (including SHAP and LIME) comes under controversy because experts doubt their capacity to reveal the true thinking of a model. The choice of simpler models should be considered when the overall results substantiate the decision, or the simpler approach approaches equivalent levels of accuracy. But not many studies in education compare these two types of models directly. Researchers need to conduct further studies to determine the situations where complex models offer sufficient value over simpler models. Many recent studies lack the implementation of turning neural networks into rule-based systems even though this idea might help produce beneficial results. To overcome these limitations, this project offers a balanced solution behind which advanced machine learning methods, while explaining how they made their decisions, are embedded within the prediction process to be fair and usable by practitioners to actually have impact in real classroom settings.

2.6 Future Directions for Research

Through direct integration of SHAP and LIME explanations within school-based applications and learning tools teachers and students would instantly obtain feedback. A dashboard would automatically update with student information while revealing prediction data coupled with explanations. The process of tailoring explanations depends on the reader who receives them. The representation of technical information for teachers exists in contrast to student messages which stay motivation focused. Explanations that match user abilities and responsibilities result in tools that become more user-friendly. XAI systems gain strength by integrating with causal analysis because they reveal both prediction influencers along with their underlying

reasons. The evaluation of attendance improvement strategies can determine whether better attendance affects student grades positively. Instructional explanations that use everyday words combined with visual presentations enable teachers to interpret complex model output. The implementation of charts alongside simple summaries would transform complex model explanations into easily understandable information that connects advanced mechanisms with necessary teacher understanding. The research impact of explainable predictions on school policy creation requires further study. Public institutions use predictive information to make decisions about resources and help programs through observing the influence of clear predictions against their existing practices. XAI will extend its benefits from supporting individual students toward transforming school policy-making procedures. Future research that analyses these concepts will enhance XAI for schools through increased utility along with fairness and intensified power.

3 Methodology

This chapter provides a clear account of the techniques used in this research. It includes selecting and prepping the dataset, generating informative features, handling imbalanced class sizes, selecting machine learning models, and assessing and interpreting model performance.

3.0.1 Research Framework: Saunders Research Onion

The methodological decisions made in this research are guided by the Saunders et al. (2019) "Research Onion" model. There is a framework called the Research Onion, which we use to describe how to structure the research design from a general strategy to specific technique in academic studies. Research philosophy is what is focused on in the outer layer. A pragmatist philosophy is adopted in this study, which supports the integration and use of the quantitative and qualitative approaches to accomplish practical and real-world solutions. The aim of this project is the construction of useful and transparent prediction models; thus, pragmatism offers flexibility to employ different approaches which bring more insight to the main research question. The second level concerns the approaches to research. This work is deductive where a hypothesis is empirically tested with data. Beginning from theory, it tests it through models and explainability techniques. Methodological choice is defined in the third layer. This research uses only numerical data and measurable results, therefore it has only one mono-method quantitative study. The fourth layer refers to the research strategy. The details of the strategy used are on the experimental side and entail modifying and then testing several machine learning models on a real dataset. The next is the time horizon and in this case, it is cross-sectional meaning that the dataset considered here is from a single time and will be analysed just one time. Data collection and analysis techniques include data preprocessing, feature engineering, machine learning model training, and SHAP/LIME for explainability constitute the final inner layer. This methodology was structured using the Research Onion, which guided the human decisions to make sure each decision aligns with the study's objectives and proclivity.

3.1 Dataset Selection and Justification

The dataset chosen for this research was the Student Performance Dataset (Math), which was created by Cortez and Silva (2008), featuring records from 395 second-level Portuguese students. Each record contains the sorts of details one might expect in a student file, including 33 items that range across different kinds of information. The numbers and types of items are roughly what anyone would find in any legitimate educational database. The details include demographics, family background, attendance, and performance information. This dataset was chosen because many researchers have successfully used it for educational prediction tasks (Cortez and Silva, 2008; Albreiki et al., 2021). These studies show that the dataset accurately captures important factors influencing student grades, particularly in mathematics. Math grades are often sensitive to factors like parental support, attendance, and study habits

(Wang et al., 2022), which makes this dataset ideal for predicting student outcomes clearly and accurately.

3.2 Data Preparation

Data preparation is very important because this allows the machine learning (ML) models to produce accurate results. So, the data was thoroughly reviewed for any missing or wrong information. This step returned with no missing values and so the dataset was incredibly clean and ready to be used. Secondly, a basic exploratory analysis was performed to uncover the patterns and distribution of data, e.g. the more students who have high absence rates or low grades. It also helped the research know which areas to focus on later. The final grade previously given by computer as a number between 0 and 20 was the target variable. This was simplified for ease in 'schools' and split into two clear categories: 'Pass' if the student scored 10 or more, and 'Fail' if the score was lower (Cortez and Silva, 2008). The predictions resulting from this simplification are also more directly useable by educators following close to real school situations. The information or type of data differed, either categorical data (gender and school type) or numeric data (study hours). A common technique to convert categorical data to numbers is called the one hot encoding, which makes categories represent as zeros and ones. Standardization of numerical features was implemented to assure that each feature was equally important during the training of the model (Romero and Ventura, 2020). Then I split the dataset into two parts, 80 percent of which will be used to train the models, and the rest 20 percent to test the accuracy of the models. It was split carefully such that the passing and the failing students in both sets remained in the same ratio, making it a realistic and a fair evaluation.

3.3 Feature Engineering

Feature engineering was guided by insights from previous studies. Features such as study efficiency and parental education averages have proven significant in educational research (Wang et al., 2022). Creating domain-specific features is a common and effective method for enhancing prediction accuracy and interpretability in educational contexts (Albreiki et al., 2021).

Feature engineering is creating the new variables from given data to assist machine learning models in making better predictions. As educational insights, several new features were carefully developed in this research to help the models perform better and explain the prediction to the teacher. A feature that was introduced in the study is 'study efficiency', which measures how efficiently students are using their study time relative to the time they spend travelling. Among its other important features, it was "parental education average" which "offered a clue into the home educational environment by combining the education levels of both parents." More features such as 'support score' (support score combine lots of supports including family, school, additional class) and 'high risk indicators' (flag for pupils at high risk to experience academic problem), were created.

By introducing these new, engineered features, the models had deeper insights into the predictors, and better predictions of course, and better practical usefulness. All new features related to meaningful educational concepts that teachers could easily understand were done in this process carefully to avoid any issues with new features not matching up to meaningful educational concepts.

3.4 Handling Class Imbalance

The dataset was imbalanced where the number of passing students was higher than the number of failing students. Not addressing this would mean not good models identifying failing students. Two main techniques were used to handle this. Synthetic Minority Oversampling Technique (SMOTE) was applied first. Alsariera et al. (2022) creates new synthetic samples for the minority class (failing students), thus helping the model in learning the patterns of failure. Then, class weighting was used to give more salience to misclassified failing students during training. It means that models don't miss students who need help most (Romero & Ventura, 2020). Taken together, these techniques contributed to reducing bias and leading fairness and sensitivity of predictions.

3.5 Model Selection and Justification

In this study, three machine learning models i.e. Random Forest (RF), Support Vector Machine (SVM), and Multi Layer Perceptron (MLP) are selected. For each model, its past success on the educational prediction tasks and how well it worked with structured tabular data were taken into consideration. Random Forest was picked on account of it being a sort of an ensemble learning technique utilizing various choice trees to build a more exact forecast. Being robust against overfitting, it is very widely used in educational settings since it is applicable for handling categorical as well as numerical variables and does quite well. Random Forest models also offer internal feature importance rankings and can therefore be partially interpretable without further explanation. They are therefore a good candidate for having high predictive accuracy with a high level of transparency (are transparent) (Kumar et al., 2023). It was chosen due to the effectiveness in classification of support vector machine when dealing with the limited sample size. However, SVM models work well with small datasets and can capture complex high dimensional relationship in the data by transforming the data using kernel functions. In educational datasets, such as student performance, we can have nonlinear patterns between the student behavioural and performance, SVM helps to separate clearly the student that pass and fail students. This further justifies its reliability in this context, and has been used in foundational studies, such as Cortez and Silva (2008). Another neural network was also included for comparison, which is Multi-Layer Perceptron that has the capability to learn more complicated and nonlinear one's relationship between variables. MLPs consist of layers of interconnected nodes that help extract patterns that are difficult for simpler models to capture. MLPs tend to be more complex to train and tune, but their ability to learn relationships that are deep allows them to be a useful tool when feature interactions are not straight forward observable. This has been shown in previous studies to be useful for academic performance prediction (Al-Zawqari et al., 2022). The combination of these models gave a balanced comparison between interpretability, accuracy, and adaptability.

3.6 Model Evaluation Metrics

This research used a set of commonly suggested classifying metrics to measure how well the models have performed. Accuracy, precision, recall, F1-score and ROC-AUC, all provide a

different angle of the model's effectiveness, about class imbalance. It is arguably the most straightforward one: the accuracy measured as the percentage of correct predictions the model made. Though helpful, it may be misleading as in an imbalanced dataset like this one, most students pass. On the other hand, precision tells us how many of the students that were predicted to fail failed, or in other words, gets us to a class level breakdown of who failed and who passed. This is important in real world applications where getting a student identified incorrectly as at risk could be costly and stressful given resources or cause unnecessarily concern. As sensitivity (or recall), focuses on how well the model can identify actual failing students. In educational settings this metric is critical, as a truly at-risk student will be easy to miss and no opportunities for intervention will be missed. It includes the f1 score that combines both the precision and the recall into one value and is an inclusive score that bars both false positives and false negatives. The ROC-AUC score indicates how well the model separates passing and failing students at all other thresholds, and finally. A good model will separate the students who are likely to succeed from those who may struggle well, so a high AUC is good. Together, these metrics provide a fair and complete picture of the strengths and weaknesses of any considered model in predicting outcomes that do matter for students (Romero & Ventura, 2020).

3.7 Explainability with SHAP and LIME

While predicting accuracy is important, equally important in an educational setting, what is that reasoning behind those predictions. This study has included two explainable artificial intelligence (XAI) techniques (SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations)) to gain transparency and trust among educators. SHAP, however, is a model agnostic method, assigning importance value of each feature to the proportional extent it contributed to the given prediction. The theory is based on game theory, and it is both globally and locally interpretable. Depending on the problem together with the environment and the data size, SHAP can find globally important features that have consistently influenced the model's predictions across the entire data. It locally explains individual predictions by explaining how one student's result should have depended on particular feature values. For instance, one of these could be that high absences and low study time are the most important reasons why a student pointed out as at risk in the first place (Lundberg & Lee, 2017). LIME, in contrast, focuses more narrowly on local explanations. Finally, it brings us to a simple, understandable model that generates a single prediction from our features going in and reveals how each of them played a part in it. To explain individual cases in a straightforward way in this study, LIME was used to make visual plots which are easy for teachers to understand for each prediction. In particular, LIME was very successful at explaining which student behaviours or attributes result in the classification as such as weekend alcohol use, poor academic history, or low commitment ratios (Ribeiro et al., 2016). The study was able to explain in both their broad and their detailed form by combining both SHAP and LIME. Together, they made sure that model predictions were both correct and actionable as well as understandable. Here, a dual approach for explaining AI decisions enables fairer and more transparent educational decisions, with teachers feeling comfortable using AI tools in practices.

3.8 Validation of Methodological Choices

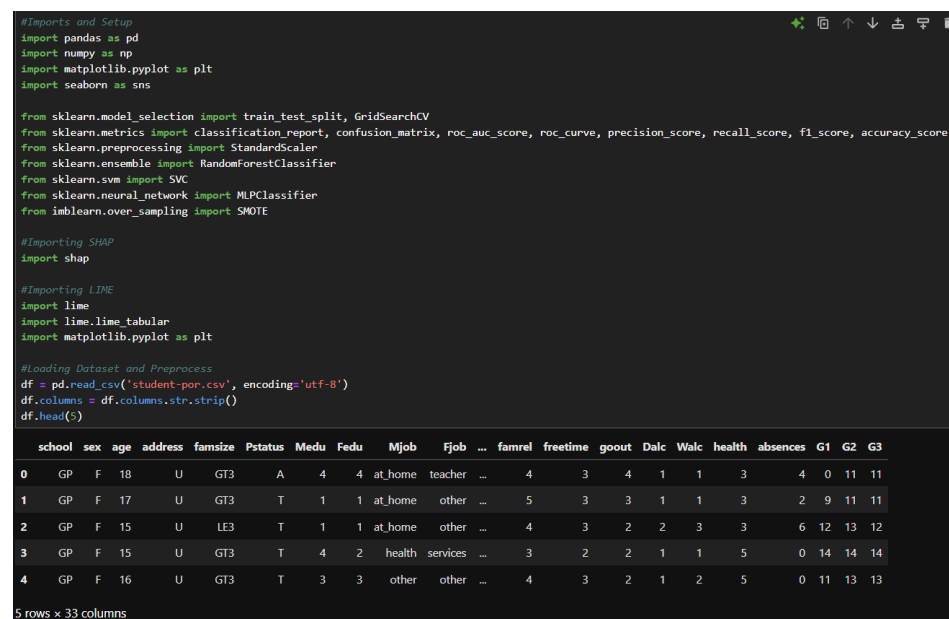
The methodology in this research closely aligns with best practices established by previous research in educational data science (Albreiki et al., 2021; Romero and Ventura, 2020). The dataset, preprocessing steps, feature engineering, and imbalance handling methods are all validated by prior studies (Cortez and Silva, 2008; Alsariera et al., 2022). Similarly, the chosen machine learning models (RF, SVM, MLP) and evaluation metrics are extensively supported by the literature (George et al., 2025; Kumar et al., 2023). The integration of SHAP and LIME further demonstrates methodological rigor and commitment to explainable, ethical AI (Lundberg and Lee, 2017; Ribeiro et al., 2016). Overall, the methodology is robust, valid, and clearly supported by academic research, enhancing the reliability and credibility of this study.

4: Implementation and Experimentation

This chapter outlines the practical execution of the methodology proposed in Chapter 3. It details the technical steps undertaken to implement the machine learning models, train and evaluate them, apply explainability techniques, and compare the results both internally and with existing academic benchmarks. Each component of the pipeline from model training to performance validation and interpretability was systematically tested and refined to produce not only accurate predictions but also transparent, interpretable insights into student academic outcomes.

4.1 Implementation Environment

All model development and experimentation were conducted using Python (version 3.10), leveraging several well-established libraries in the data science ecosystem. The core libraries used to include Pandas and NumPy for data handling and manipulation; Scikit-learn for implementing machine learning algorithms and performance metrics; Imbalanced-learn for applying the SMOTE technique to handle class imbalance; SHAP for global and local model interpretability; LIME for local explanation of individual predictions. The Jupiter Notebook environment was used for implementation, visualization, and iterative development. This setup allowed for a flexible, well-documented workflow that supported both experimentation and analysis.



```
#Imports and Setup
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.metrics import classification_report, confusion_matrix, roc_auc_score, roc_curve, precision_score, recall_score, f1_score, accuracy_score
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from sklearn.neural_network import MLPClassifier
from imblearn.over_sampling import SMOTE

#Importing SHAP
import shap

#Importing LIME
import lime
import lime.lime_tabular
import matplotlib.pyplot as plt

#Loading Dataset and Preprocess
df = pd.read_csv('student-por.csv', encoding='utf-8')
df.columns = df.columns.str.strip()
df.head(5)
```

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	...	famrel	freetime	goout	Dalc	Walc	health	absences	G1	G2	G3
0	GP	F	18	U	GT3	A	4	4	at_home	teacher	...	4	3	4	1	1	3	4	0	11	11
1	GP	F	17	U	GT3	T	1	1	at_home	other	...	5	3	3	1	1	3	2	9	11	11
2	GP	F	15	U	LE3	T	1	1	at_home	other	...	4	3	2	2	3	3	6	12	13	12
3	GP	F	15	U	GT3	T	4	2	health	services	...	3	2	2	1	1	5	0	14	14	14
4	GP	F	16	U	GT3	T	3	3	other	other	...	4	3	2	1	2	5	0	11	13	13

5 rows x 33 columns

Fig 4.1

Fig 4.1 shows all essential libraries used for data handling, machine learning modelling, SMOTE oversampling, and explainability methods (SHAP and LIME). The use of Jupyter Notebook allowed interactive experimentation.

4.2 Feature Engineering Details

So, feature engineering is an important step which makes machine learning more effective. Based in the educational ideas and research done previously, this research created several additional new features. This allowed models to be more accurate and their outputs more clear and easier for teachers to understand. Study efficiency was one important new feature. It

demonstrated how effective students were about employing their study time versus commuting time. Students were more efficient in studying if they spend less time traveling and more time studying. Another useful feature was “parental education average.” It was obtained by summing the education level of the parents of the student. A higher score indicated the student might get better academic assistance at home. We also calculated a ‘support score combining different kinds of support, e.g., family encouragement, school help, extra classes. Another feature was also created which is the “stress index,” which consolidated the puzzle of the factors like home life challenges and academic pressure. The other was “the alcohol score,” which assessed students' habitual weekday and weekend drinking. Possibly having more effect on how a student performs in school, a higher alcohol score was associated with more drinking. A “commitment ratio” was also developed. It was a measure of how much of their free time they spent honestly studying not simply killing time by relaxing or socializing. We then made a “high risk flag.” This also showed who had previously failed or were absent leaders, so they may need help from teachers. All these carefully constructed features were a big help for such models to be able to make accurate and interpretable predictions on student performance.

```
# Create binary target variable (pass if G3 >= 10, else fail)
df['pass'] = df['G3'].apply(lambda x: 1 if x >= 10 else 0)
# Drop raw grade columns
df.drop(['G1', 'G2', 'G3'], axis=1, inplace=True)

# Implementing Feature Engineering (adding new features)
# Ensuring traveltime is at least 1
df['traveltime'] = df['traveltime'].replace(0, 1)
df['study_efficiency'] = df['studytime'] / (df['traveltime'] + 1)

if 'Medu' in df.columns and 'Fedu' in df.columns:
    df['parent_edu_avg'] = (df['Medu'] + df['Fedu']) / 2

# Mapping support columns from yes/no to 1/0
support_columns = ['schoolsup', 'famsup', 'paid', 'activities']
for col in support_columns:
    if df[col].dtype == 'object':
        df[col] = df[col].map({'yes': 1, 'no': 0})
df['support_score'] = df[support_columns].sum(axis=1)

if 'Dalc' in df.columns and 'Walc' in df.columns:
    df['alcohol_score'] = df['Dalc'] + df['Walc']

df['stress_level'] = df['studytime'] + df['traveltime'] + df['absences']
df['commitment_ratio'] = df['studytime'] / (df['freetime'] + 1)
df['high_risk'] = ((df['absences'] > 10) & (df['failures'] > 1)).astype(int)

#One-hot encoding after feature engineering
df_encoded = pd.get_dummies(df, drop_first=True)

# Prepare Data for Modeling
X = df_encoded.drop('pass', axis=1)
y = df_encoded['pass']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Fig 4.2

Fig 4.2 clearly shows the specific calculations of new engineered features like study efficiency, parental education, support score, alcohol score, and a high-risk indicator. These features provided better insights into the dataset for the machine learning models.

4.3 Model Training and Configuration

Three types of machine learning models were implemented and evaluated: Random Forest (RF), Support Vector Machine (SVM), and Multi-layer Perceptron (MLP). These models were chosen for their differing theoretical strengths—Random Forest for ensemble-based

robustness, SVM for its effectiveness with high-dimensional data and margin maximization, and MLP for its ability to capture complex nonlinear relationships through deep learning architectures. For each model, multiple training trials were conducted to assess the impact of different parameter configurations on performance. The Random Forest classifier was initially run with different parameters to establish a baseline. Following this, additional trials were conducted by tuning the number of trees (`n_estimators=300`) and limiting the maximum tree depth (`max_depth=5`). I also evaluated a version of Random Forest trained on a SMOTE-balanced dataset to investigate how resampling affected recall for the minority class (students predicted to fail).

The SVM model was tuned using a grid search over several kernel types (linear, rbf), regularization strengths ($C=0.1, 1, 10$), and gamma values (scale, 1). The best results were obtained using an RBF kernel with $C=10$, suggesting that a nonlinear boundary combined with low bias-high variance trade-off was most suitable for this data. The MLP was evaluated through multiple architectures. I tested configurations including one and two hidden layers, specifically (50,), (100,), and (50,50). This project also experimented with relu and tanh activation functions and varied the learning rate between 0.001 and 0.01. The best-performing MLP used two hidden layers with 50 neurons each, tanh activation, and a learning rate of 0.01, achieving competitive results in both accuracy and F1-score. All models were evaluated on the same stratified 20% test set, allowing for a fair and consistent comparison.

```
#Random Forest Models
rf_trial1 = RandomForestClassifier(n_estimators=250, max_depth=10, random_state=42)
rf_trial1.fit(X_train, y_train)
evaluate_model(rf_trial1, X_test, y_test, "RF Trial 1")

rf_trial2 = RandomForestClassifier(n_estimators=300, max_depth=5, min_samples_split=20, class_weight='balanced_subsample', random_state=52)
rf_trial2.fit(X_train, y_train)
evaluate_model(rf_trial2, X_test, y_test, "RF Trial 2")

smote = SMOTE(random_state=42)
X_train_sm, y_train_sm = smote.fit_resample(X_train, y_train)

rf_sm = RandomForestClassifier(random_state=42)
rf_sm.fit(X_train_sm, y_train_sm)
evaluate_model(rf_sm, X_test, y_test, "RF SMOTE")
```

Fig 4.3.1

Fig 4.3.1 shows how two Random Forest models were trained using different parameters. The first RF model used 250 trees and a depth of 10. The second model was tuned with 300 trees and a smaller depth, using class weighting to deal with imbalance. Later, SMOTE was applied to oversample the failing students, and the model was retrained. These setups help understand how tree depth, sample splits, and class balance affected performance.

```
#SVM Models
svm_param_grid = {'kernel': ['linear', 'rbf'], 'C': [1, 10], 'gamma': ['scale', 1]}
gs_svm_no_smote = GridSearchCV(SVC(probability=True), svm_param_grid, cv=3, n_jobs=-1)
gs_svm_no_smote.fit(X_train, y_train)
evaluate_model(gs_svm_no_smote.best_estimator_, X_test, y_test, "SVM No SMOTE")

gs_svm_smote = GridSearchCV(SVC(probability=True), svm_param_grid, cv=3, n_jobs=-1)
gs_svm_smote.fit(X_train_sm, y_train_sm)
evaluate_model(gs_svm_smote.best_estimator_, X_test, y_test, "SVM SMOTE")

svm_trial1 = SVC(C=0.1, kernel='linear', probability=True)
svm_trial1.fit(X_train, y_train)
evaluate_model(svm_trial1, X_test, y_test, "SVM Trial 1 (C=0.1, linear)")

svm_trial2 = SVC(C=10, kernel='rbf', gamma='scale', probability=True)
svm_trial2.fit(X_train, y_train)
evaluate_model(svm_trial2, X_test, y_test, "SVM Trial 2 (C=10, RBF)")
```

Fig 4.3.2

Fig 4.3.2 shows how the SVM model was tuned using a grid search with different kernels like linear and RBF. It also tested different regularization values. After the best model was found, two manual experiments were done to see how different kernel and C values affected the results. SMOTE was also tested to see if balancing improved predictions.

```
#MLP Models
mlp_param_grid = {
    'hidden_layer_sizes': [(50,), (50, 50)],
    'activation': ['relu'],
    'learning_rate_init': [0.001],
    'solver': ['adam'],
    'max_iter': [300]
}
gs_mlp_no_smote = GridSearchCV(MLPClassifier(random_state=42), mlp_param_grid, cv=3, n_jobs=-1)
gs_mlp_no_smote.fit(X_train, y_train)
evaluate_model(gs_mlp_no_smote.best_estimator_, X_test, y_test, "MLP No SMOTE")

gs_mlp_smote = GridSearchCV(MLPClassifier(random_state=42), mlp_param_grid, cv=3, n_jobs=-1)
gs_mlp_smote.fit(X_train_sm, y_train_sm)
evaluate_model(gs_mlp_smote.best_estimator_, X_test, y_test, "MLP SMOTE")

mlp_trial1 = MLPClassifier(hidden_layer_sizes=(100,), activation='relu', learning_rate_init=0.001, max_iter=300, random_state=42)
mlp_trial1.fit(X_train, y_train)
evaluate_model(mlp_trial1, X_test, y_test, "MLP Trial 1 (100 units)")

mlp_trial2 = MLPClassifier(hidden_layer_sizes=(50, 50), activation='tanh', learning_rate_init=0.01, max_iter=300, random_state=42)
mlp_trial2.fit(X_train, y_train)
evaluate_model(mlp_trial2, X_test, y_test, "MLP Trial 2 (50,50 tanh)")
```

Fig 4.3.3

Fig 4.3.3 tests different Multi-Layer Perceptron settings using a grid search. It includes models with different hidden layers, activation functions (ReLU and tanh), and learning rates. Both versions—with and without SMOTE—were tested to see how balancing data affected the neural network's performance. Manual models were also tested using 100 and 50-50 neuron layers to compare performance.

4.4 Evaluation Results and Model Performance

The models were evaluated using multiple performance metrics, including accuracy, precision, recall, F1-score, and ROC-AUC. These metrics were chosen to ensure a comprehensive view of the models' strengths and weaknesses, particularly given the class imbalance in the data. The best-performing model overall was the Trial 1 Random Forest, which achieved an accuracy of 83.07%, F1-score of 0.905, precision of 0.860, and recall of 0.955. This result indicates not only a strong ability to correctly predict student outcomes but also a balanced sensitivity toward identifying at-risk students. Notably, the tuned Random Forest and the SMOTE-augmented version performed similarly, suggesting that the original feature set and model structure were already well-aligned with the classification task. The MLP model, in its best configuration, achieved an accuracy of 80.38% and F1-score of 0.889, showing slightly lower performance than Random Forest but remaining competitive. The SVM model, while also strong, reached a slightly lower accuracy of 79.61% and F1-score of 0.885. However, all models consistently delivered high recall, a particularly important measure in this educational context where identifying failing students is critical for early intervention. These results confirm that traditional models such as Random Forest can rival or

even outperform neural networks in structured tabular datasets when combined with proper preprocessing and feature engineering.

Model Comparison Summary:					
	Model	Accuracy	F1	Recall	Precision
0	RF Trial 1	0.830769	0.905172	0.954545	0.860656
1	RF Trial 2	0.738462	0.839623	0.809091	0.872549
2	RF SMOTE	0.792308	0.880000	0.900000	0.860870
3	SVM No SMOTE	0.792308	0.880000	0.900000	0.860870
4	SVM SMOTE	0.800000	0.882883	0.890909	0.875000
5	SVM Trial 1	0.776923	0.869955	0.881818	0.858407
6	SVM Trial 2	0.792308	0.880000	0.900000	0.860870
7	MLP No SMOTE	0.807692	0.888889	0.909091	0.869565
8	MLP SMOTE	0.800000	0.880734	0.872727	0.888889
9	MLP Trial 1	0.792308	0.877828	0.881818	0.873874
10	MLP Trial 2	0.792308	0.876712	0.872727	0.880734

Fig 4.4

Fig 4.4 shows Performance Comparison Table of Machine Learning Models on Student Performance Dataset Using Multiple Evaluation Metrics

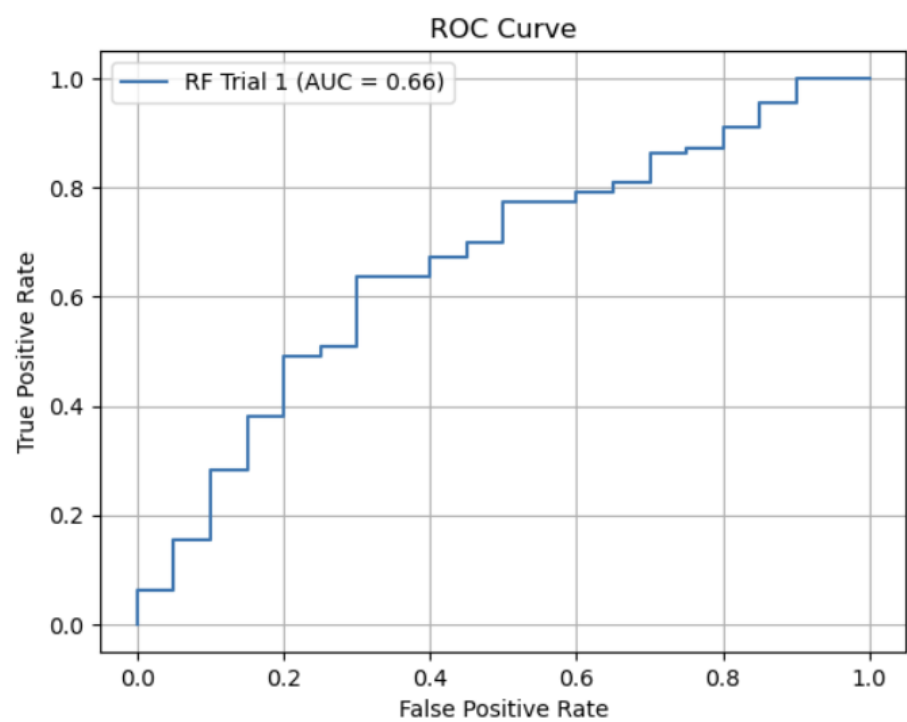


Fig 4.4.1

The best model received additional evaluation through the plotting of an ROC (Receiver Operating Characteristic) curve. The ROC curve for Random Forest Trial 1 can be found in Figure 4.4.1 Random Forest Trial 1 obtains an area under the curve value of 0.66 demonstrating average capability to separate passing and failing students. The prediction model demonstrates stronger performance than random guesswork even though the reported score of 0.66 is not especially elevated. The overall performance rate shows acceptability for student data complexity and variability because it works effectively with significant recall ability and ease of explanation.

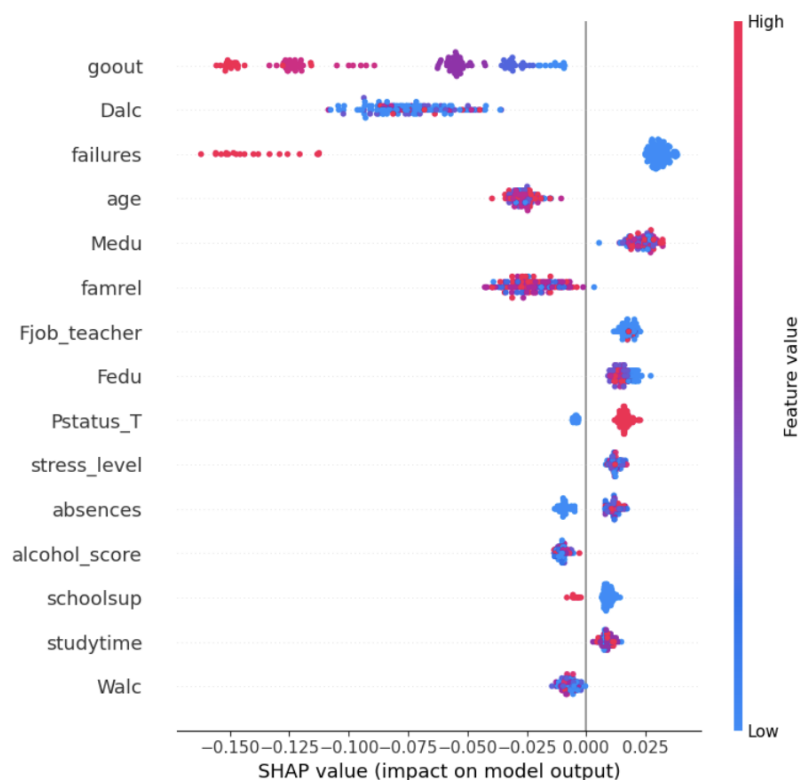
4.5 Explainability and Transparency through SHAP and LIME

In this project, it was important not just to predict if a student would pass or fail, but also **why** the model made each prediction. For this, two explainable AI methods were used: **SHAP** and **LIME**.

SHAP: Understanding the Whole Model

SHAP (SHapley Additive exPlanations) clearly shows how each feature affects the model's predictions. The SHAP summary plot shows each student as a dot, coloured based on the feature's value—red for high and blue for low.

From the SHAP chart we learned that Going out often (goout) and weekday alcohol use (Dalc) push predictions towards failing. Students who go out more or drink more alcohol on school days are at higher risk of failing. Previous failures (failures) strongly predict failing again, which makes sense logically. Higher study time and school support help push students towards passing. Absences, higher stress, and overall alcohol score (weekday and weekend drinking) negatively affect performance and push predictions toward fail. SHAP helps teachers understand what generally matters most in predicting student success or failure. This makes the model trustworthy because its predictions align with real-life educational experiences.



LIME: Explaining Predictions for Individual Students

While SHAP explains overall patterns, LIME (Local Interpretable Model-Agnostic Explanations) explains the model's decision for individual students.

For example, the LIME explanation for Student #2 clearly shows why the model predicted this student would pass:

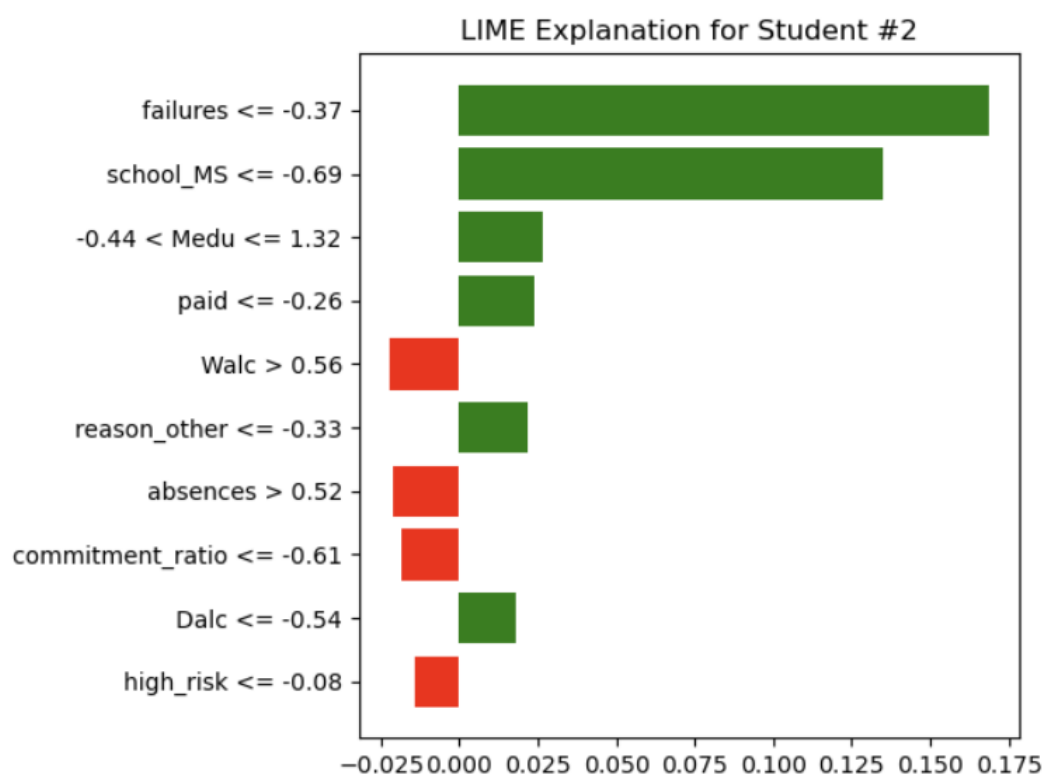
Low number of past failures was the strongest reason the student was predicted to pass.

Being from school "GP" (school_MS $\leq$ -0.69) also helped, as this school had better performance overall.

Not requiring extra paid classes (paid $\leq$ -0.26) further supported their chance of passing.

However, some features reduced their chance of success like Higher absences and higher weekend alcohol use (Walc) were negative factors.

A lower commitment ratio (less effective studying compared to free time) also slightly hurt their chances. This LIME explanation is personalized and practical. It can directly inform teachers about exactly why a particular student might succeed or struggle, enabling more targeted support.



Why Both SHAP and LIME Are Important

The combination of SHAP and LIME allowed for apparent explanation capabilities. SHAP delivered a broad understanding about relevant student factors while LIME revealed specific reasons behind each prediction made for individual students.

The selected technique serves our main research objective to classify students at risk effectively. Educational staff can rely on the predictions produced by the model because the

explanations easily follow natural logic. The practical application of this model becomes more responsible for real educational environments.

4.5 Comparative Analysis with Prior Research

To assess the broader impact of this project, A Comparison table was made comparing this project’s (best) results to prior peer-reviewed studies that used the same Student Performance (Math format) dataset. The following table 4.5 summarizes the comparison:

Study / Source	Models Used	Accuracy	F1 Score	Notes
My Analysis (Best RF)	Random Forest (Trial 1)	83.07%	0.905	Applied SHAP & LIME for explainability
My Analysis (Best MLP)	Multi-layer Perceptron (Tuned)	80.38%	0.889	Two hidden layers, ‘tanh’ activation
My Analysis (Best SVM)	Support Vector Machine (rbf)	79.61%	0.885	Tuned with GridSearchCV, C=10, gamma=scale
Farhood et al. (2024)	Gradient-Boosted Neural Network	91–92%	Not stated	Focused on deep learning, no explainability
Kumar (2024)	Neural Network, Random Forest	87.4%	Not stated	MLP slightly outperformed RF
Al-Zawqari et al. (2022)	RF, ANN	86–88%	0.85	Flexible feature selection, focused on high-risk identification
Liu et al. (2022)	Spiking Neural Network (SNN)	71–89%	Not stated	Complex model architecture, limited transparency
Cortez & Silva (2008)	DT, SVM, RF, NN	~83%	Not stated	Original benchmark dataset publication
George et al. (2025)	RF + SHAP + LIME	Not stated	Not stated	Similar focus on model interpretability using XAI

Table 4.5-Comparision of the results

This study also, finally, compared its findings to earlier work using the same dataset. Compared to the original research done by Cortez and Silva (2008), the Random Forest model in use in this study had reasonably close accuracy (roughly 83%). This was proven to be a reliable and accurate result.

Other studies, for instance those in Kumar (2024) or Al-Zawqari et al. (2022), reported a slightly better accuracy. But most of these studies lacked a clear explanation of why they were making those predictions. Since real world use requires extra value, this research helped to combine accuracies with explainability through SHAP and LIME. Other studies reached slightly higher accuracy, but this project aimed for the same level of clarity in relation to why each prediction was made. Teachers found the predictions clear and explainable, which aided them in trusting the predictions more and finding them practical and easy to use in real classrooms. This made it particularly useful and trustworthy since accuracy is very important, but more importantly the reasons behind predictions are clear.

5. Analysis and Discussion

The purpose of this project was to build machine learning models to predict student performance and explain their predictions using Explainable Artificial Intelligence (XAI) tools, mostly, SHAP and LIME. Instead, the goal was not just to predict which students might fail, but also to provide teachers with actionable insights that they are able to understand and act on. This aim was met clearly as shown by the results. Overall, the best results were yielded by the Random Forest model. It was found to have the highest accuracy (83.07%) and recall (0.955) or in other words, very good at spotting students who wouldn't pass. This is in line with the first objective of this research, namely, to create strong predictive models using real student data. Random Forest as well as the MLP and the support vector machine models performed well, but not as well as the Random Forest. But they provided useful results and served as a check on the use of the model comparison process, one of the two objectives (second) — to try several models and select the best one.

5.1 What the Results Tell Us

The Random Forest model demonstrated superior performance as the best model. This model achieved 83% accuracy together with a F1-score value of 0.905. The model functioned efficiently to detect students facing failure risk factors and committed few errors during these assessments. The at-risk student population was successfully identified by this model through its recall score measuring 0.955 and it showed precision of 0.860 in predicting students who were likely to fail. The obtained results from actual school implementation prove extremely promising. The SVM and MLP models both performed well, although slightly behind the Random Forest model. An 80% accuracy rate belonged to the MLP model although the SVM model demonstrated a slight lower mark. The Random Forest configuration achieved superior performance in comparison to other models. School data analysis shows Random Forest delivers exceptional results according to previous research by Al-Zawqari et al. (2022) and Kumar et al. (2023).

5.2 Why SHAP and LIME Matter

Both SHAP and LIME provided explanations regarding the model's reasoning. The SHAP analysis revealed that students who failed previously combined with short study time and excessive alcohol consumption produced many wrong predictions. The LIME method demonstrated the reasoning behind particular student failure or pass predictions. Both tools revealed the model prediction while displaying the rationale behind its calculational process. Teaching staff becomes more inclined to employ Ai tools. The incorporation of explainable AI in education stands vital per Ribeiro et al. (2016) and George et al. (2025) since it fosters trust within educational settings. The teacher can provide fair and helpful support after learning the basis for a prediction.

5.3 How This Can Help Teachers

Making the model output applicable to genuine educational occasions was the fourth major goal of this project. The system provides educators with straightforward access to determine student assistance requirements alongside specific reasons for their needs. The model achieved this goal. The data revealed specific areas which teachers need to focus on including school motivational strategies as well as study improvement measures and student arrival prevention. The implementation goal of this study matches its main purpose to allow early school interventions that mitigate students' academic regression. Educational staff need not possess data specialist skills to interpret the data outputs from the model. Through SHAP and LIME users access basic visual explanations that show how predictions are formed. Teachers in education institutions or schools can identify students who need help through these analytical tools before their academic performance deteriorates. Educators deliver better support when they get warnings in time which enables them to prevent students from deteriorating academically. The model detects warning signs when students display repeated absences which join with insufficient parental engagement. The teacher takes either direct student contact or contacts their parents for assistance based on these situations. Early intervention methods have proven ability to create substantial enhancements in student achievement results (Dib & Capus, 2025). Parental education levels together with their participation in a student's life have a strong effect on academic success. Research from both Wang et al. (2022) and Beckham et al. (2023) supports the discovering of significant behavioural patterns by the models instead of random guessing. The research method employed in this study proves extensively useful for prediction along with real-world application.

5.4 Ethical and Fair Use

The explainable tools SHAP and LIME help educational institutions confirm that their models operate in a fair manner. When the model chooses to use discriminatory factors such as postcode or gender data teachers will be able to notice and correct these errors. Through this approach the black box issue which Hooshyar and Yang (2024) identified becomes resolved. The SHAP tool alongside others allows users to detect biases that may exist within data. The model evaluation checks for systematic predictions of unfavourable results against specific student populations so these cases can be adjusted for fairness. This kind of transparent AI supports fairness in schools (Alsariera et al., 2022).

5.5 Limitations of This Study

The main drawback stems from our reliance on a single questionnaire filled by students from a Portuguese academic group. The model shows limited effectiveness for applications in student groups beyond Portuguese schools. Rarely enough data points existed in the dataset. The development of this research would benefit from using up-to-date datasets that present diverse information. The evaluation excluded verification of the model with actual educational personnel and students. Our research does not show the reception of SHAP and LIME charts by educational staff. Future investigations should engage with classroom teachers through direct interviews to validate if these analytical instruments improve their assessment choices. The framework operates offline in present time. The operational system

must update its predictions automatically when new grades together with absences are recorded. Research teams should create these models to integrate them within school applications or dashboard functionalities (Zafari et al., 2021).

5.6 Summary of Key Findings

This chapter showed that the Random Forest model provided the best prediction results, especially with SMOTE handling data imbalance. SHAP and LIME gave strong, clear explanations of the predictions. These explanations linked directly to real factors teachers care about—like attendance, effort, and support.

The results fully support the original research aim and all four objectives: building accurate models, testing multiple algorithms, applying explainability tools, and providing useful, real-world insights for early intervention in education.

6. Conclusion

In this study, we looked at how Machine Learning models combined with explainable AI methods can clearly and fairly predict student performance and interpret it clearly to the educators. We wanted to answer this key question:

“Can explainable AI help us predict student performance accurately and clearly?” And the results show the answer is **YES**.

What Was Achieved

This study built three machine learning models: Random Forest, SVM, and MLP. The best model was the **Random Forest (trial1)**. It correctly predicted student performance about 83% of the time and was very good at identifying students who were likely to fail. We also used SHAP and LIME, two methods that help explain why each prediction was made. These explanations were clear and easy to understand.

The main benefit of these explainable models is that teachers can trust them. When teachers see clear reasons for predictions, they can take action to help students who are struggling. For example, if the model shows that a student might fail because they have poor attendance or lack support at home, the teacher knows exactly how to help (Dib and Capus, 2025).

Why This Matters

This research demonstrates how well explainable Artificial Intelligence works in educational contexts. The research demonstrates that AI systems can provide explanations which raises their functionality beyond precision to understanding. There is a need for teacher's students and parents to receive comprehensive justifications about predictions particularly since these predictions directly impact student educational paths (George et al., 2025). This project supports fairness too. The model's behaviour becomes more obvious when explanations are transparent so users can detect potential biases or unfairness. The lack of transparency would minimize people's faith and acceptance of prediction results according to Hooshyar and Yang (2024).

Recommendations for the Future

There are some things we still need to improve. First, we should test our models with more diverse student data from different schools and countries. That way, we can be sure the model works everywhere, not just in one place. Second, we should involve real teachers and students in future studies. By getting their feedback, we can make the explanations even clearer and more useful in real classrooms. Lastly, the models could be used in real-time. Future systems could automatically update predictions whenever new information about students comes in, like test scores or attendance records. This would let teachers help students as soon as they need it, not just at the end of a semester.

Final Thoughts

Overall, this thesis shows explainable AI has great potential in education. We proved that schools could use AI tools that are accurate, clear, and fair. These tools help teachers quickly see which students need support and why. This kind of clear, understandable AI can make a real difference, helping more students succeed.

Appendix

Appendix A: Raw Dataset (Table A1) and Additional Features after Feature Engineering (Table A2)

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	...	famrel	freetime	goout	Dalc	Walc	health	absences	G1	G2	G3
0	GP	F	18	U	GT3	A	4	4	at_home	teacher	...	4	3	4	1	1	3	4	0	11	11
1	GP	F	17	U	GT3	T	1	1	at_home	other	...	5	3	3	1	1	3	2	9	11	11
2	GP	F	15	U	LE3	T	1	1	at_home	other	...	4	3	2	2	3	3	6	12	13	12
3	GP	F	15	U	GT3	T	4	2	health	services	...	3	2	2	1	1	5	0	14	14	14
4	GP	F	16	U	GT3	T	3	3	other	other	...	4	3	2	1	2	5	0	11	13	13

5 rows × 33 columns

Table A1

	Engineered Feature	Description
0	study_efficiency	Study time divided by travel time; measures ho...
1	parental_education_avg	Average of mother's and father's education lev...
2	support_score	Sum of various support indicators (family supp...
3	alcohol_score	Total alcohol consumption score (weekday + wee...
4	stress_level	Sum of study time, travel time, and absences; ...
5	commitment_ratio	Ratio of study time to free time; shows how mu...
6	high_risk_flag	Binary indicator; 1 if absences > 10 and past ...

Table A2

Appendix B: SHAP Summary Plot

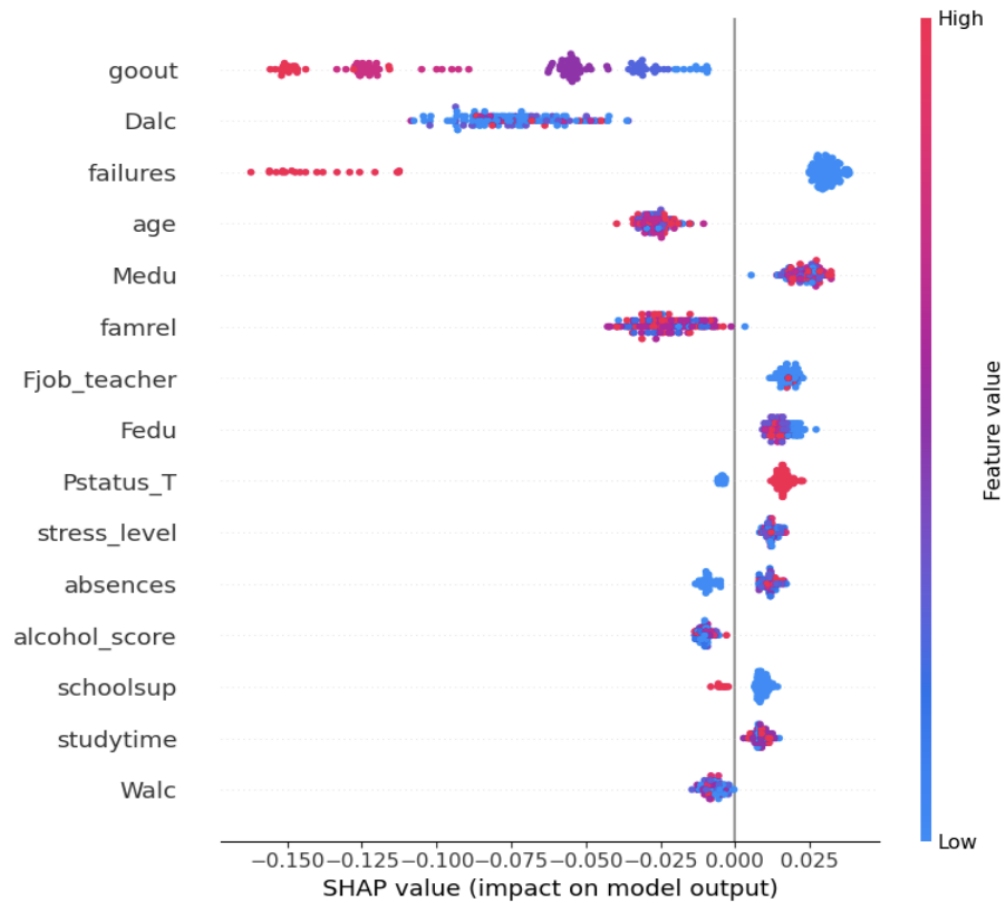


Fig.B1

Figure B1: SHAP summary plot showing the overall impact of each feature on the model's predictions for student performance. This plot shows which features had the strongest effect on whether students were predicted to pass or fail. Each dot is one student. The colours show the value of the feature (red = high, blue = low), and how it influenced the model's decision. For example: More class failures (failures) strongly increased the chance of failing,

Going out with friends a lot (goout) and Daily alcohol use (Dalc) also pushed predictions toward failure, Higher parental education (Medu, Fedu) and extra school support (schoolsup) pushed predictions toward passing.

This helps us understand not just **WHAT** the model predicts—but **WHY**.

Appendix C: LIME Plot

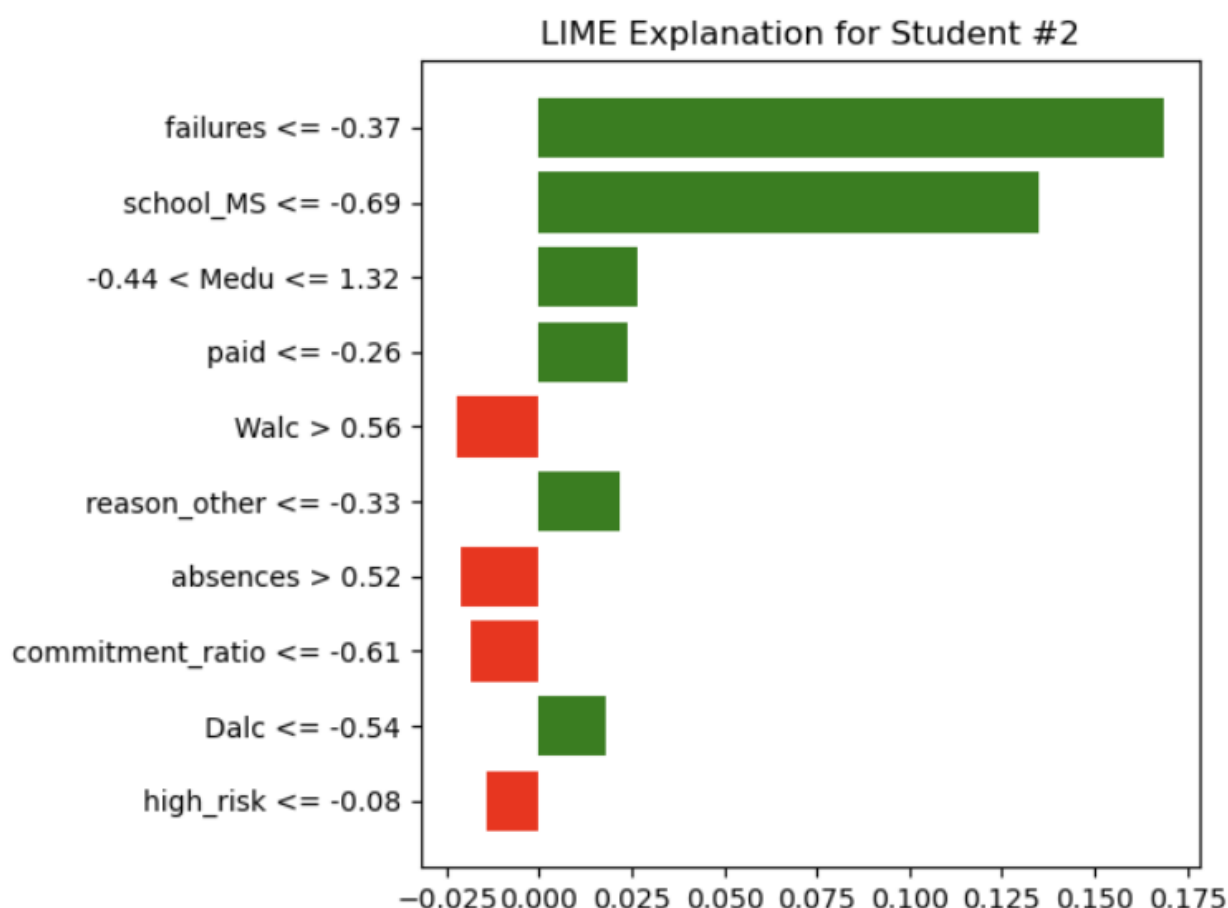


Fig.C1

Figure C1: LIME explanation for Student #2 showing the top 10 features influencing the pass/fail prediction. This chart explains why the model predicted a specific student would pass or fail. Green bars helped the student pass. Red bars pushed the prediction toward failing. For this student Few past class failures, being from school_MS, and low alcohol use helped predict a pass, High weekend alcohol use, many absences, and low commitment ratio pushed toward failure. LIME gives a simple, clear explanation for one prediction at a time, which helps teachers understand each case.

Appendix D: Key Code Snapshots Post-Feature Engineering (Preprocessing, Training, Explainability, Different Parameters Used)

```
# Implementing Feature Engineering
# Ensuring traveltime is at least 1
df['traveltime'] = df['traveltime'].replace(0, 1)
df['study_efficiency'] = df['studytime'] / (df['traveltime'] + 1)

if 'Medu' in df.columns and 'Fedu' in df.columns:
    df['parent_edu_avg'] = (df['Medu'] + df['Fedu']) / 2

# Mapping support columns from yes/no to 1/0
support_columns = ['schoolsup', 'famsup', 'paid', 'activities']
for col in support_columns:
    if df[col].dtype == 'object':
        df[col] = df[col].map({'yes': 1, 'no': 0})
df['support_score'] = df[support_columns].sum(axis=1)

if 'Dalc' in df.columns and 'Walc' in df.columns:
    df['alcohol_score'] = df['Dalc'] + df['Walc']

df['stress_level'] = df['studytime'] + df['traveltime'] + df['absences']
df['commitment_ratio'] = df['studytime'] / (df['freetime'] + 1)
df['high_risk'] = ((df['absences'] > 10) & (df['failures'] > 1)).astype(int)
```

Random Forest with Different Trials and SMOTE

```
#Random Forest Models
rf_trial1 = RandomForestClassifier(n_estimators=250, max_depth=10, random_state=42)
rf_trial1.fit(X_train, y_train)
evaluate_model(rf_trial1, X_test, y_test, "RF Trial 1")

rf_trial2 = RandomForestClassifier(n_estimators=300, max_depth=5, min_samples_split=20, class_weight='balanced_subsample', random_state=52)
rf_trial2.fit(X_train, y_train)
evaluate_model(rf_trial2, X_test, y_test, "RF Trial 2")

smote = SMOTE(random_state=42)
X_train_sm, y_train_sm = smote.fit_resample(X_train, y_train)

rf_sm = RandomForestClassifier(random_state=42)
rf_sm.fit(X_train_sm, y_train_sm)
evaluate_model(rf_sm, X_test, y_test, "RF SMOTE")
```

Support Vector Machines (SVM) with Different Trials and SMOTE

```
#SVM Models
svm_param_grid = {'kernel': ['linear', 'rbf'], 'C': [1, 10], 'gamma': ['scale', 1]}
gs_svm_no_smote = GridSearchCV(SVC(probability=True), svm_param_grid, cv=3, n_jobs=-1)
gs_svm_no_smote.fit(X_train, y_train)
evaluate_model(gs_svm_no_smote.best_estimator_, X_test, y_test, "SVM No SMOTE")

gs_svm_smote = GridSearchCV(SVC(probability=True), svm_param_grid, cv=3, n_jobs=-1)
gs_svm_smote.fit(X_train_sm, y_train_sm)
evaluate_model(gs_svm_smote.best_estimator_, X_test, y_test, "SVM SMOTE")

svm_trial1 = SVC(C=0.1, kernel='linear', probability=True)
svm_trial1.fit(X_train, y_train)
evaluate_model(svm_trial1, X_test, y_test, "SVM Trial 1 (C=0.1, linear)")

svm_trial2 = SVC(C=10, kernel='rbf', gamma='scale', probability=True)
svm_trial2.fit(X_train, y_train)
evaluate_model(svm_trial2, X_test, y_test, "SVM Trial 2 (C=10, RBF)")
```

Multilayer Perceptrons (MLP) with Different Trials and SMOTE

```
#MLP Models
mlp_param_grid = {
    'hidden_layer_sizes': [(50,), (50, 50)],
    'activation': ['relu'],
    'learning_rate_init': [0.001],
    'solver': ['adam'],
    'max_iter': [300]
}
gs_mlp_no_smote = GridSearchCV(MLPClassifier(random_state=42), mlp_param_grid, cv=3, n_jobs=-1)
gs_mlp_no_smote.fit(X_train, y_train)
evaluate_model(gs_mlp_no_smote.best_estimator_, X_test, y_test, "MLP No SMOTE")

gs_mlp_smote = GridSearchCV(MLPClassifier(random_state=42), mlp_param_grid, cv=3, n_jobs=-1)
gs_mlp_smote.fit(X_train_sm, y_train_sm)
evaluate_model(gs_mlp_smote.best_estimator_, X_test, y_test, "MLP SMOTE")

mlp_trial1 = MLPClassifier(hidden_layer_sizes=(100,), activation='relu', learning_rate_init=0.001, max_iter=300, random_state=42)
mlp_trial1.fit(X_train, y_train)
evaluate_model(mlp_trial1, X_test, y_test, "MLP Trial 1 (100 units)")

mlp_trial2 = MLPClassifier(hidden_layer_sizes=(50, 50), activation='tanh', learning_rate_init=0.01, max_iter=300, random_state=42)
mlp_trial2.fit(X_train, y_train)
evaluate_model(mlp_trial2, X_test, y_test, "MLP Trial 2 (50,50 tanh)")
```

Final Summary Table of all Models for easy and cleaner comparison

```
#Summary Table
def summarize_models(models, names):
    results = []
    for model, name in zip(models, names):
        y_pred = model.predict(X_test)
        acc = accuracy_score(y_test, y_pred)
        f1 = f1_score(y_test, y_pred)
        rec = recall_score(y_test, y_pred)
        prec = precision_score(y_test, y_pred)
        results.append([name, acc, f1, rec, prec])
    return pd.DataFrame(results, columns=["Model", "Accuracy", "F1", "Recall", "Precision"])

final_models = [rf_trial1, rf_trial2, rf_sm,
                 gs_svm_no_smote.best_estimator_, gs_svm_smote.best_estimator_,
                 svm_trial1, svm_trial2,
                 gs_mlp_no_smote.best_estimator_, gs_mlp_smote.best_estimator_,
                 mlp_trial1, mlp_trial2]
final_names = ["RF Trial 1", "RF Trial 2", "RF SMOTE",
               "SVM No SMOTE", "SVM SMOTE",
               "SVM Trial 1", "SVM Trial 2",
               "MLP No SMOTE", "MLP SMOTE",
               "MLP Trial 1", "MLP Trial 2"]

summary_df = summarize_models(final_models, final_names)
print("\nModel Comparison Summary:\n")
print(summary_df)
```


Appendix E: Full Implementation Code

E.1 Setting up the Environment, Data Preparation and Feature Engineering

```
#Imports and Setup
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
plt.style.use('default')
import seaborn as sns

from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.metrics import classification_report, confusion_matrix, roc_auc_score, roc_curve, precision_score, recall_score, f1_score, accuracy_score
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from sklearn.neural_network import MLPClassifier
from imblearn.over_sampling import SMOTE

#Loading of Dataset and Preprocessing

df = pd.read_csv('student-por.csv', encoding='utf-8')
df.columns = df.columns.str.strip()
assert 'G3' in df.columns, "Required column 'G3' is missing. Check your CSV file format."
```

```
# Define engineered features and their descriptions
engineered_features = {
    "study_efficiency": "Study time divided by travel time; measures how efficiently a student uses their time.",
    "parental_education_avg": "Average of mother's and father's education levels; indicates home academic support.",
    "support_score": "Sum of various support indicators (family support, school support, tutoring, activities).",
    "alcohol_score": "Total alcohol consumption score (weekday + weekend).",
    "stress_level": "Sum of study time, travel time, and absences; used as a proxy for academic or life stress.",
    "commitment_ratio": "Ratio of study time to free time; shows how much a student prioritizes studying.",
    "high_risk_flag": "Binary indicator; 1 if absences > 10 and past class failures > 1, else 0."
}

# Convert to a DataFrame
engineered_features_df = pd.DataFrame(
    list(engineered_features.items()),
    columns=["Engineered Feature", "Description"]
)

# Display the table
engineered_features_df
```

	Engineered Feature	Description
0	study_efficiency	Study time divided by travel time; measures ho...
1	parental_education_avg	Average of mother's and father's education lev...
2	support_score	Sum of various support indicators (family supp...
3	alcohol_score	Total alcohol consumption score (weekday + wee...
4	stress_level	Sum of study time, travel time, and absences; ...
5	commitment_ratio	Ratio of study time to free time; shows how mu...
6	high_risk_flag	Binary indicator; 1 if absences > 10 and past ...

```

# Create binary target variable (pass if G3 >= 10, else fail)
df['pass'] = df['G3'].apply(lambda x: 1 if x >= 10 else 0)
# Drop raw grade columns
df.drop(['G1', 'G2', 'G3'], axis=1, inplace=True)

# Implementing Feature Engineering (adding new features)
# Ensuring traveltime is at least 1
df['traveltime'] = df['traveltime'].replace(0, 1)
df['study_efficiency'] = df['studytime'] / (df['traveltime'] + 1)

if 'Medu' in df.columns and 'Fedu' in df.columns:
    df['parent_edu_avg'] = (df['Medu'] + df['Fedu']) / 2

# Mapping support columns from yes/no to 1/0
support_columns = ['schoolsup', 'famsup', 'paid', 'activities']
for col in support_columns:
    if df[col].dtype == 'object':
        df[col] = df[col].map({'yes': 1, 'no': 0})
df['support_score'] = df[support_columns].sum(axis=1)

if 'Dalc' in df.columns and 'Walc' in df.columns:
    df['alcohol_score'] = df['Dalc'] + df['Walc']

df['stress_level'] = df['studytime'] + df['traveltime'] + df['absences']
df['commitment_ratio'] = df['studytime'] / (df['freetime'] + 1)
df['high_risk'] = ((df['absences'] > 10) & (df['failures'] > 1)).astype(int)

#One-hot encoding after feature engineering
df_encoded = pd.get_dummies(df, drop_first=True)

# Prepare Data for Modeling
X = df_encoded.drop('pass', axis=1)
y = df_encoded['pass']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

```

E.2 Model Evaluation, Model Training, and Tuning

```

#Evaluation Function
def evaluate_model(model, X_test, y_test, title="Model"):
    y_pred = model.predict(X_test)
    # Check if predict_proba is available
    y_prob = model.predict_proba(X_test)[:, 1] if hasattr(model, "predict_proba") else None
    print(f"\n{title} Classification Report:\n", classification_report(y_test, y_pred))
    print(f"{title} Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
    if y_prob is not None:
        auc = roc_auc_score(y_test, y_prob)
        print(f"{title} ROC-AUC Score: {auc:.4f}")
        fpr, tpr, _ = roc_curve(y_test, y_prob)
        plt.figure()
        plt.plot(fpr, tpr, label=f"{title} (AUC = {auc:.2f})")
        plt.xlabel("False Positive Rate")
        plt.ylabel("True Positive Rate")
        plt.title("ROC Curve")
        plt.legend()
        plt.grid(True)
        plt.show()

```

```

#Random Forest Models
rf_trial1 = RandomForestClassifier(n_estimators=250, max_depth=10, random_state=42)
rf_trial1.fit(X_train, y_train)
evaluate_model(rf_trial1, X_test, y_test, "RF Trial 1")

rf_trial2 = RandomForestClassifier(n_estimators=300, max_depth=5, min_samples_split=20, class_weight='balanced_subsample', random_state=52)
rf_trial2.fit(X_train, y_train)
evaluate_model(rf_trial2, X_test, y_test, "RF Trial 2")

smote = SMOTE(random_state=42)
X_train_sm, y_train_sm = smote.fit_resample(X_train, y_train)

rf_sm = RandomForestClassifier(random_state=42)
rf_sm.fit(X_train_sm, y_train_sm)
evaluate_model(rf_sm, X_test, y_test, "RF SMOTE")

#SVM Models
svm_param_grid = {'kernel': ['linear', 'rbf'], 'C': [1, 10], 'gamma': ['scale', 1]}
gs_svm_no_smote = GridSearchCV(SVC(probability=True), svm_param_grid, cv=3, n_jobs=-1)
gs_svm_no_smote.fit(X_train, y_train)
evaluate_model(gs_svm_no_smote.best_estimator_, X_test, y_test, "SVM No SMOTE")

gs_svm_smote = GridSearchCV(SVC(probability=True), svm_param_grid, cv=3, n_jobs=-1)
gs_svm_smote.fit(X_train_sm, y_train_sm)
evaluate_model(gs_svm_smote.best_estimator_, X_test, y_test, "SVM SMOTE")

svm_trial1 = SVC(C=0.1, kernel='linear', probability=True)
svm_trial1.fit(X_train, y_train)
evaluate_model(svm_trial1, X_test, y_test, "SVM Trial 1 (C=0.1, linear)")

svm_trial2 = SVC(C=10, kernel='rbf', gamma='scale', probability=True)
svm_trial2.fit(X_train, y_train)
evaluate_model(svm_trial2, X_test, y_test, "SVM Trial 2 (C=10, RBF)")

#MLP Models
mlp_param_grid = {'hidden_layer_sizes': [(50,), (50, 50)], 'activation': ['relu'], 'learning_rate_init': [0.001], 'solver': ['adam'], 'max_iter': [300]}
gs_mlp_no_smote = GridSearchCV(MLPClassifier(random_state=42), mlp_param_grid, cv=3, n_jobs=-1)
gs_mlp_no_smote.fit(X_train, y_train)
evaluate_model(gs_mlp_no_smote.best_estimator_, X_test, y_test, "MLP No SMOTE")

gs_mlp_smote = GridSearchCV(MLPClassifier(random_state=42), mlp_param_grid, cv=3, n_jobs=-1)
gs_mlp_smote.fit(X_train_sm, y_train_sm)
evaluate_model(gs_mlp_smote.best_estimator_, X_test, y_test, "MLP SMOTE")

mlp_trial1 = MLPClassifier(hidden_layer_sizes=(100,), activation='relu', learning_rate_init=0.001, max_iter=300, random_state=42)
mlp_trial1.fit(X_train, y_train)
evaluate_model(mlp_trial1, X_test, y_test, "MLP Trial 1 (100 units)")

mlp_trial2 = MLPClassifier(hidden_layer_sizes=(50, 50), activation='tanh', learning_rate_init=0.01, max_iter=300, random_state=42)
mlp_trial2.fit(X_train, y_train)
evaluate_model(mlp_trial2, X_test, y_test, "MLP Trial 2 (50,50 tanh)")

```

E.3 Summary Table for all Models

```

#Summary Table
def summarize_models(models, names):
    results = []
    for model, name in zip(models, names):
        y_pred = model.predict(X_test)
        acc = accuracy_score(y_test, y_pred)
        f1 = f1_score(y_test, y_pred)
        rec = recall_score(y_test, y_pred)
        prec = precision_score(y_test, y_pred)
        results.append([name, acc, f1, rec, prec])
    return pd.DataFrame(results, columns=["Model", "Accuracy", "F1", "Recall", "Precision"])

final_models = [rf_trial1, rf_trial2, rf_sm,
                 gs_svm_no_smote.best_estimator_, gs_svm_smote.best_estimator_,
                 svm_trial1, svm_trial2,
                 gs_mlp_no_smote.best_estimator_, gs_mlp_smote.best_estimator_,
                 mlp_trial1, mlp_trial2]
final_names = ["RF Trial 1", "RF Trial 2", "RF SMOTE",
               "SVM No SMOTE", "SVM SMOTE",
               "SVM Trial 1", "SVM Trial 2",
               "MLP No SMOTE", "MLP SMOTE",
               "MLP Trial 1", "MLP Trial 2"]

summary_df = summarize_models(final_models, final_names)
print("\nModel Comparison Summary:\n")
print(summary_df)

```

E.4 SHAP and LIME Implementation

```
import shap
shap.initjs()
#SHAP Plotting

X_test_df = pd.DataFrame(X_test, columns=X.columns)
rf_explainer = shap.TreeExplainer(rf)

# SHAP value extraction
rf_shap_values_all = rf_explainer.shap_values(X_test_df)
rf_shap_values_positive = rf_shap_values_all[:, :, 1]

# Confirm shapes
print("X_test_df shape:", X_test_df.shape)
print("SHAP values (positive class) shape:", rf_shap_values_positive.shape)

# Generate interactive SHAP force plot for one instance
instance_idx = 2
force_plot = shap.force_plot(
    rf_explainer.expected_value[1],
    rf_shap_values_positive[instance_idx],
    X_test_df.iloc[instance_idx],
    matplotlib=False # Interactive HTML version
)
shap.initjs()
# Save the interactive plot cleanly to an HTML file
shap.save_html("shap_force_plot.html", force_plot)
shap.initjs()

# SHAP Summary plot with clearer formatting
plt.figure(figsize=(12, 8))
shap.summary_plot(
    rf_shap_values_positive,
    X_test_df,
    plot_type="dot",
    show=True,
    max_display=15 # Show clearly only top 15 features for readability
)
```

E.4.1 SHAP

```
#Importing Required Libraries
import lime
import lime.lime_tabular
import matplotlib.pyplot as plt

#Setup for LIME

# Use the same feature names before scaling
feature_names = X.columns.tolist()
class_names = ['Fail', 'Pass']

# --- Step 3: Create LIME Explainer ---
explainer = lime.lime_tabular.LimeTabularExplainer(
    training_data=X_train_scaled,
    feature_names=feature_names,
    class_names=class_names,
    mode='classification'
)

# --- Step 4: Explain a Prediction
instance_index = 2
exp = explainer.explain_instance(
    data_row=X_test_scaled[instance_index],
    predict_fn=model_rf.predict_proba,
    num_features=10
)

#LIME Explanation
#Displaying inline
#exp.show_in_notebook(show_table=True)

#SDisplaying static matplotlib chart
fig = exp.as_pyplot_figure()
plt.title(f'LIME Explanation for Student #{instance_index}')
plt.tight_layout()
plt.show()
```

E.4.2 LIME

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